

Sneha A. Karnik, CSM®

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EDUCATION

North Carolina State University

Master of Industrial/Engineering Management

Aug 2022 – Dec 2023

University of Mumbai

Bachelor of Engineering in Electronics Engineering

Aug 2018 – May 2021

TECHNICAL SKILLS

- **Certifications:** Certified Scrum Master (CSM), Lean Six Sigma Green Belt, IBM Design Thinking
- **Software Tools:** MySQL, Tableau, Power BI, Minitab, JIRA, Confluence, SAS, JMP, Visio, Proteus, Altium, MS Office, FlexSim, AutoCAD
- **Core Competencies:** KPI, DMAIC, FMEA, Kaizen, 5S, Quality Assurance & Control, Root Cause Analysis, Process Improvement & Documentation, Data Analysis & Visualization, Statistical Process Control, Production Planning & Control, Agile Methodology, Technical Program Management, OEE, Gauge R&R, Process Capability

WORK EXPERIENCE

Process Engineer Intern, Tesla

Sept 2023 – Dec 2023

- Led a multifaceted project, applying the **8D methodology** to identify and rectify process inefficiencies, implemented a new process flow and a **\$230,000** reduction in scrap costs.
- Achieved a **33% reduction** in scrap of **non-conforming parts** through the development and implementation of an improved scrapping rework process.
- Rebalanced a production line to optimize manufacturing process, resulting in **15%** rise in the capacity & throughput of each station, and an **Overall Equipment Effectiveness (OEE)** of **23 units per hour**.
- Eliminated task dependencies and created a new process flow incorporating parallel operations, resulting in a reduction of **cycle time** by **3 minutes**.
- Conducted in-depth **exploratory data analysis**, revised control specification limits for testing equipment, and employed **Tableau** to develop **dashboards** for the daily monitoring of testing equipment performance.
- Implemented mechanical fixtures to enhance the efficiency of fastening tasks, thereby preventing potential scrap worth **\$5500** per shift.
- Proactively engaged in **line escalations**, providing timely and effective solutions as required.

Industrial Engineer, Castal Technologies

June 2021– Jan 2022

- Created **NPI project documentation** like checklists, inspection summaries, progress updates, drawings, specifications, and designed a **process flow diagram** using **Visio** for improved visualization.
- Enhanced operations by conducting workload assessments and implementing automated sensing solutions, leading to a **12% increase in productivity** and a **10% reduction in total packing time**.
- Evaluated production operations and facility layout to define efficient manufacturing methods, equipment, and workstation setups, achieving a **25% cycle time reduction** and descending manual material handling.

ACADEMIC PROJECTS

Hepatitis B Immunization Coverage Analysis | SQL, MS Excel, Power BI, JMP, Visio

- Identified low immunization countries by extracting data using **SQL** and analyzed using ANOVA in **JMP** which helped in addressing challenges, and further visualized this data on **PowerBI** dashboards for improving traceability of vaccination coverage in affected countries.
- Created flowcharts to increase process transparency by tracking continuous improvement activities by collaborating with UNICEF to enhance the immunization coverage in the affected regions.
- Designed **Tableau** dashboards to illustrate **KPIs** on immunization patterns, illiteracy, and sentiment analysis.
- Achieved a **35%** increase in vaccine coverage in low immunization countries through an effective communication strategy.

Making Zebulon City Smart | Kanban, Lean Methodology, Risk Management, Documentation

- Organized weekly **brainstorming** sessions with the IBM project team to discuss and refine plans, making necessary updates to the **flowcharts** as needed.
- Conducted project **FMEAs**, identified and mitigated **10** potential risks and updated stakeholders accordingly.

Real Sales Construction | Project Priority Matrix, Work Breakdown Structure, Stakeholder Analysis, Change Management

- Transformed a warehouse into a flourishing restaurant, employing a precise project plan with Gantt chart timelines, **resource allocation**, and **cost estimations**.
- Reduced project expense by **18%** through precise **cost estimates** and efficient **budget management**, while fostering strong collaboration with key **stakeholders**.